Scientific writing for infectious disease modelling

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Today's learning objective

- Think about ways to communicate the value your paper
- Paper sections, flows and structures
- Scientist/modeller philosophy in writing
- Writing tips

Why do we need writing skills?

- Academic life revolves around writing
  - Papers
  - Grants
  - Job hunting
  - Admin work
- Writing is not the *primary* value of research but helps it to be delivered most effectively

What constitutes “good writing”?

- Logical flow
- Communication skill
- Precision/honesty
- English skill

Is Akira qualified to talk about scientific writing?

- You judge 😜

- Wrote manuscripts accepted in:
  - Science
  - PNAS
  - Lancet
  - Nature Health

- At least some experience in writing for broad scientific audience

Disclaimer

- Many of the today's contents are based on my own experience, philosophy and values
- I believe they are reasonable and generally well accepted, but don't assume this is the only approach
- Especially your supervisor may have a different view, then you might want to weigh different views…
- There are also many other good scientific writing resources (particularly around language styles and structures): eg Stanford Coursera

Today's menu

- "How to sell your research"— intro/cover letter/grant writing
- Logical and clear writing rules and tips: methods/results/discussion
- Modelling specific considerations
- Other writing tips

Introduction

- What are the functions of Introduction section?

Introduction

- Set the scene
- Share background knowledge required to read the paper
- Literature review
- Motivate the study
- Outline the study

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↓

- Prepare the reader to read the body of the paper
- Convince them that your study is important and worth reading

Good research ↔ Good introduction

- Clear, logical and well motivated introduction section suggests research is also designed/conducted well
- If you find it challenging to write it or it reads cluttered… maybe you need to take time to re-assess/re-articulate the overall concept of your study
- What's the core value of your research and why?

Motivation: tell a story

- Motivation for the study “to pitch”
  -  “Because my supervisor asked me…” “MOH wants it…” “I just had data…”
  -  “This hasn’t been done” “Interesting idea” “Improves understanding”
  -  “This research improves people’s health”

- How to frame your research in relation to benefit to society (not just academic society) is a critical skill
- Eventual tone is up to audience (journal) but good to think about the final goal of your research output

Key ingredients for Intro section

- Context (“set the scene”)
- Issues
- Current status
- Research gap
- What you do and how it solves the issues

Example (Endo et al. 2022)

- Mpox has been historically driven by zoonotic spillover followed by limited human-to-human transmission, but in 2022 suddenly it started rapid international outbreak through sexual contacts among MSM (Context)
- What caused such a radical shift in epidemiology remains unknown (Issues)
- Some says viral evolution, others say cessation of smallpox vaccine program; no reliable evidence (Current status/research gap)
- We used branching process model to show that differences in sexual behaviours across population groups alone can explain the shift without requiring novel hypotheses (what you do and how it solves the issues)

Ensure flow

- Don’t just present collection of facts one after another and let the reader connect the dots; you do it.

- *XX is a respiratory disease. No specific treatment exists and mortality is high among infants. Newly developed vaccine ZZ can confer good protection against infection and is approved for 5 years and older. This study aims to propose school age children as the primary target for vaccination.*

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- *XX is a respiratory disease. No specific treatment exists and mortality is high among infants. Newly developed vaccine ZZ can confer good protection against infection and is approved for 5 years and older. This study aims to propose school age children as the primary target for vaccination.*
  - Each sentence is in isolation and no clear links between them
  - Why the aim of the study?

Ensure flow

- XX is a respiratory disease with a high mortality among infants. As no specific treatment exists, prevention through efficacious vaccines has been a long-standing public health priority. Newly developed vaccine ZZ can confer good protection against infection but is approved only for 5 years and older. This means that infants, who bear the highest disease burden, could not be benefited from direct protection. However, ZZ could still offer indirect protection for infants; as older siblings are typical source of infection for infants, controlling transmission among older siblings, particularly at schools, is expected to reduce infants' exposure. This study aims to propose school age children as the primary target for vaccination.

Eliminate leaps of logic

- A, therefore B, therefore C…
- Ideal introduction facilitates and accompanies reader's thought process
- Don't "allow" reader to counterargue ("wait, why C not D?")

- *Vaccine-derived immunity against measles was previously considered to be lifelong but recent studies suggested it can wane after 20+ years (Ok).*
  → Populations whose population immunity is mostly vaccine-derived may face the risk of measles outbreak (Ok).
  → We conducted a survey to assess the willingness to receive an MMR booster dose among elderly in Singapore (Wait).

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  - Is Singapore's immunity waned enough to warrant intervention?
  - Is MMR booster the most viable option to address waning immunity?

Eliminate leaps of logic

- You may write statements with leaps when implicit logical links that you take for granted are not shared with or presented to readers
- "As older siblings are typical source of infection for infants, controlling transmission among older siblings, particularly at schools, is expected to reduce infants' exposure." This sentence actually has a leap!

Eliminate leaps of logic

- You may write statements with leaps when implicit logical links that you take for granted are not shared with or presented to readers
- "As older siblings are typical source of infection for infants, controlling transmission among older siblings, particularly at schools, is expected to reduce infants' exposure." This sentence actually has a leap!
- Missing element: "transmission among older siblings often take place at schools"
- Maybe too obvious for you, readers will accept it if they share the same context, otherwise they will stumble

How can we write a good intro?

- Do research whose value is clearest to you
- Know your audience
- Before writing, design the structure and check the logical flow
- Make sure you are understood by readers who have no idea about your research
- Proofread yourself, and ask others to proofread too

Cover letter

- Your pitch to *editor* (not reviewer or reader)
- Similar to intro, but slightly different: **why is your study worth space?**
  - What's the impact?
  - Why now?
  - Match with the readership?
- There are things that you can't write in paper but can in cover letters
  - Can use sensational tones (if not excessively)
  - Relate to timely events (WHO declarations, political events, etc)
- Don't repeat abstract and waste the opportunity to pitch!

Grant writing

- Your pitch to *funder, reviewer,* and *members of the society*
- Academic value is still important, but value to the society becomes more important
- Can you convince reviewer who may be outside of your field that your research is worth investment?
  - How you digest your research for lay audience
  - Let the reviewer relate to what your research would achieve

Pitch to reviewers outside of your field

- Eye tracking technology assisted by video recognition allows inference of a person's attention in real time. Certain motor execution tasks require a sufficiently high level of attention, below which interventions including interruption of the task should be in place to prevent undesirable consequences. This study aims to establish a method to quantify and assess individuals' attention level during tasks for monitoring and, where necessary, automated interruption.

- This may read fine, but is it convincing enough?
- What changes would you make?

Pitch to reviewers outside of your field

- Eye tracking technology assisted by video recognition allows inference of a person's attention in real time. Certain motor execution tasks require a sufficiently high level of attention, below which interventions including interruption of the task should be in place to prevent undesirable consequences. For example, identifying and alerting distracted drivers could save lives that would otherwise be lost to accidents. This study aims to establish a method to quantify and assess individuals' attention level during tasks for monitoring and, where necessary, automated interruption.

- Sometimes examples are much more powerful than logic
- Especially when you're pitching to nonexperts/ those from outside your field

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Questions?



Methods

- Critical section to ensure **reproducibility**: ie readers should be able to follow exactly what you did from the description, which requires clarity and specific details
- Nowadays methods section alone isn't sufficient for modelling studies and often specific model setups are sent to SI
- Ensuring 100% reproducibility is almost impossible: that's why publishing code helps

Typical structure of methods section

- Data source: this helps the readers to assess how much the results are data driven vs assumption driven
- Model: at least provide an overview of the structure, what it does and core assumptions made
- Parameter estimation/fitting: data and likelihood (+ prior) used, parameters calibrated, optimization method, how fitting was assessed
- (Outputs): Any results you derived using the calibrated model, e.g. impact of interventions, cost effectiveness, forecasts…

Results

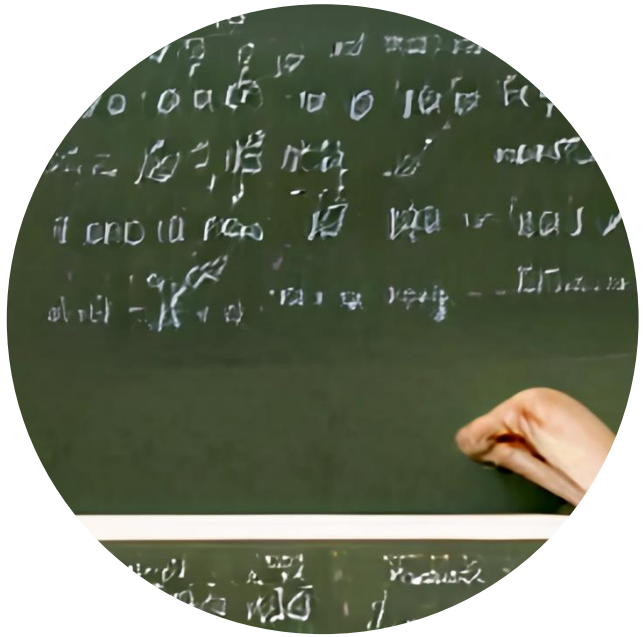
- You may feel the least creative section but you could still embed your narrative
- Don’t just overload readers with numbers! Results are usually supposed to be separated from discussion, but still you can organize the structure and flow such that it facilitates interpretation
- My personal preference is, instead of opening it by “*Fig 1 shows the estimated Rt values in city X…”,* first summarise what the results imply “Estimated Rt in city X showed a stable trend… (Fig 1).”

Figures

- Make it self-explanatory as possible: sometimes readers will see figures without looking at the main text
- Figure caption: don’t make it “necessary”, yet make it “sufficient” to understand the figure
  - If one can’t mostly understand fig without caption, it’s a bad fig
  - If one can’t fully understand fig even with caption, it’s a bad caption
- Avoid acronyms as much as possible
- Use friendly colour coding for colour vision difference
- Start y axis from 0!

Discussion

- First para: summarise the study aim and key results
- Next few paras: interpretations, comparison with previous studies, impact/implications
- Limitations: be thorough and humble. I'd frown on papers skipping obvious key limitations and only list "cliché": there may be underreporting!



Writing philosophy & Tips

Information density

- How much concrete/specific info is embedded in a text
- Minimise “empty” sentences sounding as if you’re saying something but aren’t
  - *”This study provides holistic understanding of the complex dynamics of dengue transmission.”*
  - *“Our tool helps to inform optimal control strategies against future pandemics.”*
  - *“We believe The Lancet would be the ideal platform that attracts broad attention from the global health readership ☺”*
  - Does it add more info? If not, maybe don’t waste word count?
- Alternatively:
  - *This study explains the multi-serotype dengue dynamics driven by cross reacting immunity.*
  - *Our tool estimates the optimal allocation of vaccines within supply and cost constraints.*

Be humble as a modeller

- "All models are wrong"—as such, your modelling results are never REAL results
  - "In our model simulation, children contributed most to flu transmission": Ok
  - "Children was suggested to have contributed most to flu transmission": Ok
  - "Children contributed most to flu transmission": Not Ok!!
  - Fine to use definitive language for mathematical facts or modelled entities, not real-world
- Think about how much of your results are informed by data
  - Your estimate of R_{child} is higher than R_{adults}: Did data have this signal, or because you imposed specific mixing pattern? What if contact matrix was inaccurate?
  - This assessment of "confidence" requires skill, and most readers won't see it. So be honest
  - If you feel data/model is weak, consider some form of validation
- "Model explains/fits the data well" doesn't indicate your model is plausible
  - Did you really exclude other possibilities which may explain the data even better?

Use conjunctions effectively but not excessively (especially adversatives)

All models are wrong. However, they can be useful when used with their caveats in mind. However, sometimes model outputs are oversold as if they are solid predictions. Guidelines are in development, however, to prevent such miscommunication.

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Therefore/Thus require logical links

- Therefore/thus are not a fancy version of “and”: they indicate a logical link, *therefore* logical link should be there
- When I see them, I’ll read that section with care to get the logic. I’ll get confused if it isn’t there, or has leaps—which makes the paper look weak
- So don’t use them if you don’t intend to demonstrate clear logical links

- *Secondary attack rates were substantially higher among household contacts than in other close-contact settings. Therefore, interventions to reduce household transmission can be, if designed properly, an effective epidemic control strategy.*
- *There have been fewer reports of mpox with clear growing trend recently in Africa; WHO thus lifted the declaration of public health emergency.*

By default, different terms = different things

- Sometimes you might feel like avoiding repeating the same phrase
- Rephrasing terminology is risky in scientific writing, as readers may well assume that you say it differently for a reason
- Don’t fear repetition; better to sound dull than to confuse

By default, different terms = different things

- After social distancing was implemented, the mean contact rate among participants substantially decreased. Over the study period, the mean contact rate was strongly correlated with the intensity of the intervention at that time. This suggests that the effective reproduction number was lowered by the policy.

- The estimated Rt values varied across the four regions. In region A, transmission was almost 3-fold compared with other regions throughout the study period.

Loosely defined phrases: don't assume readers know how you define them in your text

- Scientific terms with multiple usage (including misnomers)
  - Incubation period / Asymptomatic infections / Index case / XX rates
  - Spillover / Data augmentation
- Vague phrases
  - Level / Approach / Strategy / Scenario
  - Outbreak / Transmissibility

- Common languages (sometimes clearly defined in a certain context)
  - Infant / juvenile / adult / elderly / young / old

Readers have read only what they have read

- Texts are (usually) read from start to finish. Only you know what will come next
  - Explain/define terms, notions, notations, etc. when they first appear
  - Give an overall picture first, explain details later
  - Don’t require knowledge that first appears later
    - “Just accept this fact for now, it’s explained in the next section”: this doesn’t work!

- But some people read in their own order (eg Discussion first), so making sections as standalone as possible may also help: e.g. spell out abbreviations and

Difficult to parse

- Too long / nested / unbalanced / ambiguous structures
- If readers need effort to parse, it's already failure
- Sometimes readers can parse ambiguously structured sentence just because they managed to guess your intention; i.e. they can read because they know what they will read, which is NOT what scientific writing should aim for

- Split, simplify, and reorganise the sentences

Difficult to parse

- *Policymakers need to assess the additional benefit of the strictly targeted vaccination campaign against RSV with catch-up administration of monoclonal antibody against the potentially increased logistical burden that come with the selective operation.*

Difficult to parse

- *Policymakers need to assess [the additional benefit of (the strictly targeted vaccination campaign against RSV) with (catch-up administration of monoclonal antibody)] against [(the potentially increased logistical burden) that come with (the selective operation)].*

Difficult to parse

- *Policymakers need to assess the additional benefit of the strictly targeted vaccination campaign against RSV with catch-up administration of monoclonal antibody against the potentially increased logistical burden that come with the selective operation.*

- *The strictly targeted vaccination campaign against RSV combined with catch-up administration of monoclonal antibody may provide additional benefit. However, policymakers need to assess the benefit against the potentially increased logistical burden associated with the selective nature of its operation.*

This/that/it/they/which/… what does it refer to?

- *We estimated the vaccine-induced reduction in influenza deaths from the baseline projection, which we assumed to be constant regardless of the product used in the vaccination campaign.*

- If there's any chance of ambiguity, spell out
  - In a long sentence, there may be multiple candidates
  - It looks clear to you because you (only you) know what it is!
  - When in doubt, err on the safe side

Implicit comparator/reference

Phrases that only make sense with comparator/reference

- “Additional”: to WHAT?
- “Increase/decrease”: from WHAT?
- “Relatively”: compared with WHAT?
- “Proportion”: of WHAT?

Always good to specify what is comparator/reference if possible; otherwise, make sure it is uniquely identifiable from the context

Distinguish between

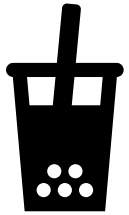
- Facts vs (mainstream vs personal) Opinions
  - *"XX is considered to be…" "XX is known to be…"*: who says that? If it's an establish fact you should be able to cite
- Direct observation vs Estimates
  - "Risk" is it a direct measure (eg proportion) or indirectly estimated?
- Causation vs Correlation
  - "*Intervention was estimated to have reduced Rt by 10%*" …Really?

Other phrases that I personally avoid

- **in terms of / in light of / in relation to**: unless the meaning is clear from context (usually it's not)
- **novel approach**: if it's really novel, readers will know even if you don't say it loud
- **XX has not been done / studied**: unless you are sure it's never been done (I often miss literature so just don't take the risk)
- **utilise / employ / deploy / leverage [methods]**: just a bit bombastic when you can just say "use"
- **elucidate / explore / investigate**: ditto, maybe just "studied" or "analysed"?

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Can you do me a favour for next year?

- I’m hoping to evolve this session to better reflect what students tend to struggle with in writing
- If OK, can I use your cover letters to understand common challenges and maybe use some as examples in the sessions for next year onwards (of course anonymously)?
- If you don’t agree, add “opt out from reuse” or something in your letter at the end so I know. Thank you!

Next ~~week~~ month

- 3 weeks off! No quizzes in the meantime. Maybe start assignment?
- Topic moves to inference; likelihood-based estimation to start off with

